EM algorithm: Derivation for GMM

Expected value of the complete log-likelihood

Let $\theta = (\pi_1, ..., \pi_{K-1}, \mu_1, ..., \mu_K, \sigma_1^2, ..., \sigma_K^2)$.

$$\begin{aligned} p(X_1, ...X_n, Z_1, ..., Z_n|\theta) &= \prod_{i=1}^{n} p(X_i|Z_i, \theta)p(Z_i|\theta) \\ &= \prod_{i=1}^{n}\prod_{j=1}^{K}[p(X_i|Z_i = j, \theta)p(Z_i = j|\theta)]^{I_{(Z_i=j)}}, \\ \log[p(X_1, ...X_n, Z_1, ..., Z_n|\theta)] &= \sum_{i=1}^{n}\sum_{j=1}^{K} I_{(Z_i=j)}[\log p(X_i|Z_i = j, \theta) + \log p(Z_i = j|\theta)]. \end{aligned}$$

$$\begin{aligned} Q(\theta, \theta^0) &= E_{Z|X,\theta^0}[\log p(X_1, ...X_n, Z_1, ..., Z_n|\theta)] \\ &= \sum_{i=1}^{n}\sum_{j=1}^{K} p(Z_i = j|X_i, \theta^0)[\log p(X_i|Z_i = j, \theta) + \log p(Z_i = j|\theta)] \\ &= \sum_{i=1}^{n}\sum_{j=1}^{K} p_{ij}^0[-\log\sqrt{2\pi\sigma_j^2} - \frac{(x_i - \mu_j)^2}{2\sigma_j^2} + \log\pi_j], \end{aligned}$$

where $p_{ij}^0 = p(Z_i = j|X_i, \theta^0)$ which is the current estimate of the conditional probability of observation i belonging to component j, given the parameter values θ^0.

E-step

For $j \in \{1, 2, ..., K-1\}$,

$$\begin{aligned} p_{ij}^0 = p(Z_i = j|X_i, \theta^0) &= \frac{p(Z_i = j, X_i|\theta^0)}{p(X_i|\theta^0)} \\ &= \frac{p(X_i|Z_i = j, \theta^0)p(Z_i = j|\theta^0)}{\sum_{k=1}^{K} p(X_i|Z_i = k, \theta^0)p(Z_i = k|\theta^0)} \\ &= \frac{\frac{1}{\sqrt{2\pi(\sigma_j^2)^0}}\exp\left\{-\frac{(X_i-\mu_j^0)^2}{2(\sigma_j^2)^0}\right\}\pi_j^0}{\sum_{k=1}^{K}\frac{1}{\sqrt{2\pi(\sigma_k^2)^0}}\exp\left\{-\frac{(X_i-\mu_k^0)^2}{2(\sigma_k^2)^0}\right\}\pi_k^0}, \end{aligned}$$

where $\pi_K^0 = 1 - (\pi_1^0 + \pi_2^0 + ... + \pi_{K-1}^0)$.

M-step

To find $\hat{\pi}_j$
For $j \in \{1, 2, ..., K-1\}$,

$$\begin{aligned} \frac{\partial Q(\theta, \theta^0)}{\partial \pi_j} &= \sum_{i=1}^{n}\Big[\frac{p_{ij}^0}{\pi_j} - \frac{p_{iK}^0}{1 - (\pi_1 + \pi_2 + ... + \pi_{K-1})}\Big] \\ &= \frac{\pi_K\sum_{i=1}^{n} p_{ij}^0 - \pi_j\sum_{i=1}^{n} p_{iK}^0}{\pi_j\pi_K} = 0. \end{aligned}$$

Hence, for $j \in \{1, 2, ..., K-1\}$,

$$\pi_K\sum_{i=1}^{n} p_{ij}^0 = \pi_j\sum_{i=1}^{n} p_{iK}^0. \tag{1}$$

Take the sum of these $K-1$ equations,

$$\pi_K \sum_{i=1}^{n} \sum_{j=1}^{K-1} p_{ij}^0 = (1-\pi_K) \sum_{i=1}^{n} p_{iK}^0.$$

Moving $\pi_K \sum_{i=1}^{n} p_{iK}^0$ to the left hand side,

$$\pi_K \sum_{i=1}^{n} 1 = \sum_{i=1}^{n} p_{iK}^0.$$

Hence,

$$\pi_K = \frac{1}{n} \sum_{i=1}^{n} p_{iK}^0.$$

Then by equation (1),

$$\hat{\pi}_j = \frac{1}{\sum_{i=1}^{n} p_{iK}^0} \pi_K \sum_{i=1}^{n} p_{ij}^0 = \frac{1}{n} \sum_{i=1}^{n} p_{ij}^0.$$

To find $\hat{\mu}_j$
For $j \in \{1, 2, ..., K\}$,

$$\begin{aligned} \frac{\partial Q(\theta, \theta^0)}{\partial \mu_j} &= \sum_{i=1}^{n} p_{ij}^0 \left[\frac{X_i - \mu_i}{\sigma_j^2} \right] \\ &= \frac{1}{\sigma_j^2} \left[\sum_{i=1}^{n} p_{ij}^0 X_i - \mu_i \sum_{i=1}^{n} p_{ij}^0 \right] = 0. \end{aligned}$$

Hence,

$$\hat{\mu}_j = \frac{\sum_{i=1}^{n} p_{ij}^0 X_i}{\sum_{i=1}^{n} p_{ij}^0}.$$

To find $\hat{\sigma}_j^2$
For $j \in \{1, 2, ..., K\}$,

$$\begin{aligned} \frac{\partial Q(\theta, \theta^0)}{\partial \sigma_j^2} &= \sum_{i=1}^{n} p_{ij}^0 \left[-\frac{1}{2\sigma_j^2} + \frac{(X_i - \mu_j)^2}{2(\sigma_j^2)^2} \right] \\ &= -\frac{1}{2(\sigma_j^2)^2} \left[\sigma_j^2 \sum_{i=1}^{n} p_{ij}^0 - \sum_{i=1}^{n} p_{ij}^0 (X_i - \mu_j)^2 \right] = 0. \end{aligned}$$

Hence,

$$\hat{\sigma}_j^2 = \frac{\sum_{i=1}^{n} p_{ij}^0 (X_i - \mu_j)^2}{\sum_{i=1}^{n} p_{ij}^0}.$$

An alternative way to find $\hat{\pi}_j$

Since we have a constraint $\sum_{k=1}^{K} \pi_k = 1$, we use Lagrange multipliers and consider the Lagrangian

$$L = Q(\theta, \theta^0) + \lambda \left(\sum_{k=1}^{K} \pi_k - 1 \right).$$

We can find the maxima of $Q(\theta, \theta^0)$ constrained to $\sum_{k=1}^{K} \pi_k = 1$ by solving $\frac{\partial L}{\partial \theta} = 0$, where π_K is explicitly included in θ.
For $j \in \{1, 2, ..., K\}$,

$$\begin{aligned} \frac{\partial L}{\partial \pi_j} &= \sum_{i=1}^{n} \frac{p_{ij}^0}{\pi_j} + \lambda = 0 \\ \pi_j &= -\frac{1}{\lambda} \sum_{i=1}^{n} p_{ij}^0. \end{aligned} \tag{2}$$

The sum of these K equations gives

$$1 = \sum_{j=1}^{K} \pi_j = -\frac{1}{\lambda} \sum_{i=1}^{n} \sum_{j=1}^{K} p_{ij}^0 = -\frac{1}{\lambda} \sum_{i=1}^{n} 1 = -\frac{n}{\lambda}.$$

Hence, $\lambda = -n$. Substituting this into (2) gives

$$\hat{\pi}_j = \frac{1}{n} \sum_{i=1}^{n} p_{ij}^0.$$

Special case: $K = 2$

If $K = 2$, it is much easier to find $\hat{\pi}_j$.

$$\begin{aligned} \frac{\partial Q(\theta, \theta^0)}{\partial \pi_1} &= \sum_{i=1}^{n} \Big[\frac{p(Z_i = 1|X_i, \theta^0)}{\pi_1} - \frac{p(Z_i = 2|X_i, \theta^0)}{1 - \pi_1} \Big] \\ &= \frac{(1 - \pi_1) \sum_{i=1}^{n} p(Z_i = 1|X_i, \theta^0) - \pi_1 \sum_{i=1}^{n} p(Z_i = 2|X_i, \theta^0)}{\pi_1 (1 - \pi_1)} = 0. \end{aligned}$$

Hence,

$$(1 - \pi_1) \sum_{i=1}^{n} p(Z_i = 1|X_i, \theta^0) - \pi_1 \sum_{i=1}^{n} p(Z_i = 2|X_i, \theta^0) = 0.$$

Rearrange and solve for π_1,

$$\hat{\pi}_1 = \frac{1}{n} \sum_{i=1}^{n} p(Z_i = 1|X_i, \theta^0).$$

Then $\hat{\pi}_2 = 1 - \hat{\pi}_1 = \frac{1}{n} \sum_{i=1}^{n} p(Z_i = 2|X_i, \theta^0)$.